

Go2 RL Locomotion

Training a Unitree Go2 Quadruped to Walk in NVIDIA Isaac Lab

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Abstract

This project trains a Unitree Go2 quadruped robot to walk using deep reinforcement learning inside NVIDIA Isaac Lab. A Proximal Policy Optimization (PPO) agent is trained across thousands of GPU-parallel environments on NYU's Greene HPC cluster, learning a robust locomotion policy directly from simulated interaction.

The policy is shaped through a carefully engineered reward structure and hardened with domain randomization so that the learned gait transfers reliably from simulation toward real hardware. The full pipeline is reproducible end-to-end on SLURM and produces training metrics, checkpoints, and rollout videos of the trained gait.

Technical Highlights

- **Massively parallel training:** PPO run across thousands of simultaneous GPU environments for high sample throughput.
- **Principled reward design:** velocity tracking, posture stabilization, foot-clearance, slip minimization, and action-smoothness terms, with phase-coupled oscillators encouraging a natural, rhythmic gait.
- **Robustness by design:** domain randomization over link masses, ground friction, and external push disturbances builds a policy resilient to real-world variation.
- **Reproducible HPC pipeline:** the whole training run is driven by SLURM batch jobs on NYU Greene, with TensorBoard logging and automatic checkpointing.
- **Rich diagnostics:** rollout videos and metric curves make it easy to compare baseline, randomized, and refined policies.

Technologies Used

- **Languages:** Python
- **Frameworks:** PyTorch, NVIDIA Isaac Lab / Isaac Sim
- **Algorithm:** Proximal Policy Optimization (PPO)

- **Compute:** CUDA GPUs, NYU Greene HPC cluster, SLURM job scheduler
- **Tooling:** TensorBoard, checkpointing, automated rollout-video capture
- **Platform:** Unitree Go2 12-DOF quadruped model

Outcome

The trained policy produces a smooth, stable quadruped gait and remains robust under randomized dynamics and external disturbances, demonstrating a complete, reproducible sim-to-real reinforcement-learning workflow for legged locomotion.